Indian Institute of Technology Dharwad

॥ सा विद्या या विमुक्तये ॥
ಭಾ.ತಂ.ಸಂ. ಧಾರವಾಡ
भा. प्रौ. सं. धारवाड
I.I.T. DHARWAD

Information Brochure

Ph.D. Admissions

Spring 2026-27

Version 1.2

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Section I

General Information on Admission Process

A. SCHEDULE OF Ph.D. ADMISSION

All the updates regarding the admission process/important dates will be available on the institute website under section Admissions >> Ph.D. All potential candidates are requested to keep visiting the institute website regularly for updates.
Any query related to admission may send through email to **pgadmissions@iitdh.ac.in**

B. APPLICATION CATEGORIES & FINANCIAL SUPPORT

IIT Dharwad admits Ph.D. candidates under different admission categories for which details are given below. Availability of respective admission categories in a particular department can be found in the department-specific section of this document.

B.1. Teaching Assistantship (TA)

Funded by the Ministry of Education (MoE), Government of India, the assistantship is available for a maximum period of 5 years or up to the thesis defense, whichever is earlier. At present, the monthly rate of assistantship is ₹37,000/- for the first 2 years and an enhanced rate of ₹42,000/- for the remaining 3 years and HRA as per rules. The continuation of the assistantship will be subject to the satisfactory performance of the academic, research performance as well as duties assigned by the Departments. The employees (with or without pay) of any organization are not eligible for admission under this category. If they want admission under this category, they have to resign and submit a relieving letter from their employer before joining the program. The reservation to various social categories is applicable as per the Govt. of India norms.

Students getting assistantships from the Institute may join projects sponsored by external agencies and obtain corresponding fellowships in lieu of TA ship.

B.2. Fellowship Awardee (FA)

The financial support under this category is provided by various Government / Semi-Government schemes (for example, CSIR, UGC, DAE, DST, DBT, NBHM, PMRF, Visvesvaraya, etc.) and some other organizations. A valid Junior Research fellowship (JRF) award letter from the Government / Semi-Government agencies (e.g. CSIR / UGC / DAE / DST / DBT / NBHM / (confirmed) DST INSPIRE, etc.) is required for the execution of this fellowship.

The amount, duration, disbursement and continuation of the fellowship, HRA, etc. will be as per norms of the awarding agency or IIT Dharwad, as deemed fit.

B.3. Project Assistantship (PA)

Funded from projects sponsored by industries and government funding agencies. Under this category, candidates will be paid fellowship as per the rules & regulations of the governing project.

B.4. Externally Sponsored Ph.D. (EX)

The candidates employed in recognized R&D organizations with at least two years of work experience in the relevant area and desirous of pursuing Ph.D. program while in employment may apply for admission as external candidates**. Persons working in colleges/universities are not eligible under this category.** After fulfilling the coursework requirement at the Institute, these candidates will be allowed to register for Ph.D. with a Supervisor (internal) from the Institute and a Co-supervisor (external) from their parent organization where they will be doing the research work. The admissions are based on the following norms:

1. The competence of these candidates will be assessed along with the regular candidates.
2. Along with the application, the candidate should submit a Sponsorship Certificate (Appendix A) from the organization in which he/she is employed giving an undertaking that the candidate would be released from the normal duties to fulfill the coursework requirement (and qualifier examination, if applicable). The certificate should also provide details of facilities relevant to the research program and available to the candidate.
3. The candidate is required to be at the Institute as a full-time student for the coursework (and qualifier examination, if applicable) of his/her Ph.D. Program. The coursework requirement is likely to be a period of 1-2 semesters. Depending on the student's background and the program requirements, an additional semester may be needed to complete the coursework/qualifier examination.
4. To promote interaction between the internal supervisor and external co-supervisor, meetings between them should be arranged at least once a year in the Institute or in the sponsoring organization.
5. The Ph.D. registration of an external candidate would be reviewed at the end of each year from the date of registration in terms of his progress in courses/seminars / approved research program by a Research Progress Committee (RPC) nominated by the concerned Department Postgraduate Committee (DPGC).
6. At the time of joining the program, the students will have to produce a “Relieving certificate” from his / her employer that he/she has been fully relieved from normal duties during the semester(s) to complete the course work and other academic work at IIT Dharwad.

B.5. Institute Staff (IS) Ph.D.

Existing permanent employees of IIT Dharwad can be admitted under the category Institute Staff subject to fulfillment of conditions mentioned in the Ph.D. Rule Book. The competence of these candidates will be assessed along with the regular candidates.

B.6. College Teacher (CT) Ph.D.

Teachers from the AICTE/UGC approved colleges in the vicinity of IIT Dharwad with a minimum of 2 years of work experience can apply in this category. After fulfilling the coursework requirement at the Institute, these candidates will be allowed to register for Ph.D. The admissions are based on the following norms:

1. The competence of these candidates will be assessed along with the regular candidates.
2. Along with the application, the candidate should submit a no objection certificate (NOC) from the organization in which he/she is employed giving an undertaking that the candidate would be released from the normal duties to fulfill the coursework requirement (and qualifier examination, if applicable). The certificate should also provide details of facilities relevant to the research program and available to the candidate.

3. The candidate is required to be at the Institute as a full-time student for the coursework (and qualifier examination, if applicable) of his/her Ph.D. Program. The coursework requirement is likely to be a period of 1-2 semesters. Depending on the student's background and the program requirements, an additional semester may be needed to complete the coursework/qualifier examination.
4. The Ph.D. registration of a CT candidate would be reviewed at the end of each year from the date of registration in terms of his progress in courses/seminars / approved research program by a Research Progress Committee (RPC) nominated by the concerned Department Postgraduate Committee (DPGC).
5. At the time of joining the program, the students will have to produce a "Relieving certificate" from his / her employer that he/she has been fully relieved from normal duties during the semester(s) to complete the course work and other academic work at IIT Dharwad.

B.7. Quality Improvement Programme (QIP) Ph.D.

The Government of India launched the QIP in 1970. One of the main objectives of the programme is to upgrade the expertise and capabilities of the faculty members of the degree and diploma level institutions in the country. The programme is implemented and monitored by All India Council for Technical Education.

The eligibility for admission under QIP Ph.D. scheme is as follows: -
* Full time regular / permanent faculty members of AICTE approved Degree / Diploma Level institutes having
(a) Three-year teaching experience at polytechnic/graduate level institutes.
(b) A Master degree in the appropriate branch.

Candidates seeking admission to the Ph.D. program under the Quality Improvement Program (QIP) at IIT Dharwad must apply through the AICTE QIP portal.

All the details regarding QIP can be found in https://qip.aicte.gov.in/home

B.8. Visvesvaraya PhD Fellowship (VF)

Visvesvaraya PhD Scheme is an institutional scheme where PhD seats are allocated to the institutions, and the institutions enroll PhD candidates on these seats following their admission procedures.

Fellowship: Rs. 38,750 monthly in 1st& 2nd year and @Rs. 43,750 per month in 3rd,4th and 5th years of PhD. (Support till PhD completion or 5 years whichever is earlier)
Research Contingency Grant Support: An amount of Rs. 1,20,000/Year/Full-time PhD candidate for support duration of PhD candidate

Other benefits are - support for attending International Conference and visiting a lab abroad, HRA benefits (if the hostel is not available)
(Reference link - https://phd.digitalindiacorporation.in/eligibility-criteria)

C. GENERAL GUIDELINES FOR APPLYING ONLINE

1. Please read all the instructions given in the brochure carefully before filling in the application form.

2. The link for the online application will be updated on the website.

3. Click on Register/ Signup, and provide your details. Please note that the phone number and email ID that you provide while registering should be valid till the admission process is completed. All the communication will be through this email ID.

4. After registration, you will get an email with your User ID/Username and along with a link to set your password. Set a new password by clicking on the link. Note down your User ID and password for future reference.

5. Login using your User ID and password. >> Fill out the form step by step >> Keep all the documents ready in soft copy >> Pay the application fee through the IIT Dh. e-pay facility. https://epay.iitdh.ac.in/Payment/AdvanceFee > User ID and Date of Birth to be filled. [User ID will be generated after Register/Signup the application] >> Note down e-pay transaction No.>> Fill the remaining details in the CIMS application>> Note down the submission ID for future reference and you will get a mail confirmation of the application.

6. The application fee is as follows:

Gen/Gen (EWS)/OBC/ all other candidates	₹ 200/-
Female/SC/ST/PwD category candidates	₹ 100/-

7. **The Application Form without valid online payment details will not be considered. Application FEE once paid is Non-Refundable.**

8. **An applicant needs to pay the application fee only once irrespective of the number of departments the applicant is applying to.**

9. Applicants may find it convenient to keep following information ready while filling the application form online (whichever relevant).

- Passport size photograph [50 kb to 200 kb, JPEG/PNG formats only]
- Identity Proof (Adhaar Card, Voter ID Card, Passport etc.)
- Educational qualification details from secondary school onwards.
- NET/GATE/Other relevant qualifying examination details (if applicable).
- Fellowship related documents, e.g., JRF Award Letter, (if applicable).
- Research Proposal (For applicants in the Department of HEART)
- List of fellowships/ awards (if applicable).
- Details of the Publications (if applicable).
- Sponsorship Letter and CV of co-supervisor if you are applying under ‘EX’ category. (merged as a single PDF)(if applicable).

10. Every department may offer only some modes of financial support/application categories. Please ensure that you meet the eligibility criteria for that category/ mode. Refer to the brochure for further details.

11. Choose your Department and eligibility for the Application carefully after going

through the brochure. Once you proceed to the next page, these options can not be modified.

12. Please verify all your details before clicking on the 'Submit' button. Applications once submitted can not be modified. **Your application is liable to be rejected for incomplete/ false information.**

13. You can apply for multiple departments. Separate applications should be submitted for each department.

14. **Keep checking the institute website and your emails regularly for any communication from the institute regarding the selection process. It is the candidate's responsibility to be aware of the schedule of various events related to the admission process.**

15. The Shortlisted candidates' list will be uploaded on the institute website as per the schedule given.

16. Candidates (if) called for written test / interview should bring with them Photo ID Card, Printed Copy of Online Application Form, Photocopies of Academic Transcripts, Degree Certificates & Experience Certificates, social category certificate (if applicable), PwD Certificate (if applicable), Thesis/Dissertation/Report/Publications and all other relevant documents.

17. **Please note that the candidates (if selected) should be able to produce all relevant documents within a short period of notice. If the documents are not produced within the deadline, the admission is liable to be cancelled.**

18. The candidates who secure admission under OBC-NCL and EWS categories are required to produce the OBC-NCL/ EWS certificate issued **after 01 April 2026**. The certificate should be produced at the time of admission. The category certificate must be in the **format applicable to Central Government institutions**, and the state Government formats will not be accepted under any circumstances.

19. The SC/ST/PwD candidates who attend interviews **in person** at IIT Dharwad are eligible for **Travel Fare reimbursement i.e. AC-III Tier in train**. Candidates can also travel by public bus (taxi, chartered bus, etc. not admissible) or air, however the claim will be limited to AC-III tier fare if higher in other modes. The tickets should only be booked through the IRCTC website.

D. INFORMATION PERTAINING TO HOSTELS

About IIT Dharwad	Kindly visit the website https://www.iitdh.ac.in/ for available facilities
Hostel Room Allocation (on sharing basis)	You will be allotted a room in the hostel & the room key will be handed over on your arrival at the Institute. Each room will accommodate roughly two/four students (depending on the prevailing conditions).
Are hostel rooms furnished	Each student will be provided with a cot, chair & study table and wardrobe. Students can purchase mattress/bedding, bucket, etc. locally. Arrangements will be made for on-campus shopping for these items.
Possession of motorized Vehicle	Bicycles are permitted on campus. Motorized vehicle permission is as per Institute norms as amended from time to time.
Climatic conditions	The weather at Dharwad is pleasant throughout the year. Generally, it will be raining in the months of June to September and weather will be windy and cold during the months of October to January. It is suggested that you carry protective clothing accordingly.

E. FEES, DEPOSITS & HOSTEL RENT

The tentative fee applicable for admission to the Ph.D. program is provided below for reference purposes.

E.1. Details of Applicable Fee for Admission:

Academic Components (Amount in ₹)		First Semester	
Sl.No.	**PARTICULARS**	**GEN/EWS/OBC-NCL**	**SC/ST/ PwD**
A	**One Time Fee**		
A1	Admission	300	300
A2	Academic Service Charges	1,200	1,200
A3	Convocation Fee	3,000	3,000
A4	Alumni Lifetime Membership	2,000	2,000
	Total (A1 to A4)	**6,500**	**6,500**
B	**Refundable Deposit**		
B1	Library Deposits	1,000	1,000
B2	Institute Deposit	5,000	5,000
	Total (B1 to B2)	**6,000**	**6,000**
C	**Per Semester Fee**		
C1	Registration Fee	1,500	1,500
C2#	Tuition Fee	2,500	-
C3	Examination Fee	2,000	2,000
C4	Library Fee	500	500
	Total (C1 to C4)	**6,500**	**4,000**
I.	**Total (Academic Components)**	**19,000**	**16,500**
D	**Per Semester Fee**		
D1	Gymkhana Fee	3,000	3,000
D2	Medical and Wellness fee	2,500	2,500
D3	Student Benevolent Fund	1,000	1,000
D4	Hostel Establishment and Service Charges	2,500	2,500
D5	Mess Establishment and Service Charges	1,500	1,500
D6	Students' Activity Establishment Charges	2,000	2,000
D7	Hostel Room Rent*	1,000	1,000
D8	Electricity & water charges*	2,500	2,500
D9	Mess Advance*	24,500	24,500
	Total (D1 to D9)	**40,500**	**40,500**
E	Medical Insurance Premium (Annually)	1,500	1,500
F	**Refundable Deposit**		
F1	Hostel Deposit	5,000	5,000
II.Total (Non-Academic Components) (D+E+F)		**47,000**	**47,000**
TOTAL (I+II)		**66,000**	**63,500**

Note:

a. #IIT Dharwad reserves the right to revise the fee in future.

b. ##All the SC/ST/Divyangjan students are exempted from paying tuition fees.

c. All the students are required to pay the entire fee as per the fee circular. The students who wish to reside outside may be permitted to do so with the permission of the institute only after reporting to the campus. The respective fee components related to the hostel and mess advance will be reimbursed later with the approval of the competent authority.

d. The tuition fee for the EX category will be ₹ 25,000/-.

Section II

Department-Specific Information

F. DEPARTMENT OF CHEMICAL ENGINEERING (CE)

F.1. Eligibility for admission

F.1.1. Qualifying Degree

- B.Tech./B.E degree or equivalent degree for a direct Ph.D. program with a valid GATE score in any discipline/paper. However, GATE score can be waived for B.E./B.Tech. students of CFTI (Centrally Funded Technical Institute), having CPI/CGPA 8 (on scale of 10) or above.
- M.Tech./M.E./ M.S.(Engg.)/M.Sc or equivalent degree in Chemical Engineering or any related stream. M.Tech. (or equivalent degree) students are exempted from GATE qualification for TA category.
- M.Sc. students must have a valid GATE score/ CSIR/UGC-NET-LS for TA category.
- Junior Research Fellowship (JRF) of CSIR/UGC/DST INSPIRE/DBT/MHRD/ICMR or any other relevant funding agencies is mandatory for FA category.
- Candidates in the final year of their qualifying degree may be considered for admission to the Ph.D. programme, provided they graduate before the registration date specified by the Institute.
- Project Staff-Teaching/Research Assistantship through Project (PS-TAP/RAP): The students admitted in this type are funded from the specific sponsored projects. They are expected to assist in the academic/administrative work of the Institute.

F.1.2. Minimum score in the qualifying degree

For General/EWS/OBC-NCL category candidates and/or for candidates where no concession in academic performance is called for, the eligibility criteria in the qualifying degree (M.Tech./M.E./M.S. (Engg.)/ M.Sc):

1. A minimum of 60% marks (without round off) in aggregate, OR,

2. A minimum Cumulative Grade Point Average (CGPA) or Cumulative Performance Index (CPI) of 6.0 on the scale of 0-10; with corresponding proportional requirements when the scales are other than on 0-10, (for example, 4.8 on a scale of 0-8).

For SC/ST category candidates and differently abled candidates (PwD), a relaxation of 5% (or CPI/CGPA of 0.5 on the scale of 0-10) in the qualifying degree is applicable.

F.1.3. Eligibility of applicants in the final phase of getting the qualifying degree

Students who are in the final phase of receiving the above-mentioned qualifying degree and who are likely to graduate before commencement of Spring 2026-27 semester of IIT Dharwad are also eligible to apply. However, if offered, the admission to those candidates would be provisional. To join an academic program at IIT Dharwad, such candidates need to furnish necessary documents regarding completion of the degree on the date of joining mentioned in Section A above. They need to meet the criteria specified in the section above

considering updated score in the qualifying degree. In the meanwhile, the aggregate academic performance announced by the respective university till the last date for submission mentioned in section A should be used to determine eligibility for application and same to be reported in the online application. Externally sponsored Ph.D. (EX) applicants may also apply for the admission. The candidates will be shortlisted as per the rules and regulations applicable to EX category. The detailed guidelines and requirements are mentioned in Section B.4

F.2. Modality of selection process

Our Ph.D. selection process is highly competitive, and only eligible applicants will be permitted to the interview. The process is designed to evaluate the applicant's research potential, academic background, and motivation for pursuing a Ph.D. in Chemical Engineering. The interview will be conducted online and will involve an informal discussion about the applicant's research interests and academic background. This will also provide an opportunity for applicants to interact with our faculty members and gain a deeper understanding of our research programs.

F.3. Syllabus

- For candidates from Chemical Engineering and related fields: GATE syllabus
- For candidates other than Chemical Engineering: GATE/NET syllabus relevant to their respective fields.

F.4. Focus area of research

The following research areas are floated in the Department of Chemical Engineering for the Ph.D. program this semester. Applicants have to choose at least one of these research areas and fill in the application form. We encourage all eligible applicants to apply for our program and look forward to welcoming the most promising candidates to our department.

1. **Energy Storage and Conversion:**
   a) Electrochemical and photoelectrochemical water splitting, electrochemical CO_2 reduction, NH_3 synthesis, and methane oxidation, batteries, supercapacitors, fuel cells, design of cathode materials, and Solid-State Electrolytes Biocatalytic approach for CO_2 capture and Conversion, Biofuel production.
   b) Battery thermal management and heat transfer, Battery electrochemical and multiphysics modeling, Coupled electrochemical-thermal analysis of batteries and fuel cells

2. **Advanced Materials:**
   a) Nanomaterials (Magnetic, metallic, polymeric, semiconductor, composites, and carbon-based), Smart and Functional Materials (Shape memory alloys and polymers, self-healing materials, phase change materials, piezoelectric and magnetostrictive materials), Biomaterials (Bioinspired and biocompatible materials, drug delivery systems, antimicrobial coatings, and implants), Energy and Sustainable Materials (Materials for energy storage-batteries, supercapacitors, photovoltaic materials, hydrogen storage materials, thermoelectric and piezoelectric energy harvesting), Advanced Polymers and Composites (High-performance polymer composites, conductive polymers, biodegradable and sustainable polymers, lightweight aerospace and automotive composites), Optical Materials (Plasmonic and photonic materials, advanced coatings for optical applications, 2D materials for optoelectronic devices), Ceramics and

High-Performance Materials (Advanced refractory materials, transparent ceramics for optical applications, ceramic-matrix composites).
b) **AI-driven predictive modeling for new materials**, machine learning for materials discovery, modeling, and simulation of materials formation, Gas separation Membranes: Synthesis and modelling of ultra-thin membranes for gas separation. Development of sensors for methane. Development of catalysts for hydrogen generation.

3. **Computational Fluid Dynamics (CFD) and Machine Learning (ML):** Microplastics Separation from Industrial wastewater; Slurry Transport in Industrial pipelines; Thermal Management of Li-ion Battery with CFD & Machine Learning (ML); Gas–liquid two-phase flow patterns in a polymer–electrolyte membrane (PEM) fuel cells; CFD coupled with population balance modeling (CFD-PBM) for gas-solid (i.e., fluidization) and liquid-solid (i.e., slurry) flows; Microfluidics; Droplet based separation of rare earths; ML-integrated droplet microfluidics; Droplet Coalescence, splitting and mixing; Janus droplet formation and dynamics; Taylor bubble dynamics in a concentric annulus, Machine learning for advanced energy materialS, AI for materials discovery.

4. **Computer Aided Soft Matter Design and Engineering:** First Principles and AI aided Drug and Material Design, Drug and Material Formulation, Controlled Drug Release and Drug Delivery Systems, Molecular Simulation, Enzymes for Pharma API Synthesis, Protein and Polymeric Membrane based Separation.

5. **Sustainable Waste Valorization, Separation, Resource Recovery & Environmental Processes**
   a) Conversion of complex waste streams including plastics, biomass, electronic waste, and spent batteries into value-added chemicals, fuels, and materials through catalytic and thermochemical processes with studies on reaction kinetics and product distribution; Selective metal recovery through hydrometallurgical routes and liquid–liquid extraction (LLE) supported by thermodynamic modeling of metal nitrate systems; Electrochemical separation processes and ionic transport modeling for resource recovery from complex streams; Design of sustainable processes emphasizing resource recovery and zero liquid discharge (ZLD); Integration of experimental studies with process modeling and data-driven approaches including machine learning for analysis and prediction of chemical processes.
   b) Metal extraction and Rare Earth Element (REE) recovery, Battery recycling and critical-material recovery, Plasma-assisted treatment of industrial wastewater, Micro- and nanobubble-based separation and slurry treatment, Catalytic and thermochemical valorization of plastics, biomass, and e-waste into chemicals, fuels, and materials, including reaction kinetics and product distribution.

6. **Process modelling, simulation, and analysis :** Modelling simulation and optimization of green chemical processes (e.g., waste utilization, increasing efficiency of existing production processes, etc.), and analysis of sustainability and viability of processes in view of current scenarios.

7. **Advanced Materials, Microreactors & Process Intensification:** Synthesis of advanced functional materials, nanoparticles, and supraparticles, Continuous-flow and microreactor technologies, Biodiesel and sustainable fuel synthesis, Microfluidic platforms for materials synthesis and process intensification

F.5. Interview Instructions

If you have been shortlisted for the interview process, you will be invited to participate in an online interview. The online interview will be conducted via Google Meet, and you will receive detailed instructions and a link to access the interview prior to the scheduled date. Here are some important guidelines to follow during the online interview:

1. Technical requirements: Make sure you have a reliable internet connection and a functioning webcam and microphone on your computer or device. You should also test your equipment and internet connection in advance to ensure that they are working properly.
2. Log in to the video conferencing software at least 10 minutes prior to the scheduled interview time.
3. During the interview, speak clearly, maintain good eye contact, and be professional in your demeanor. Remember to listen carefully to the interviewer's questions and respond thoughtfully.
4. Find a quiet and distraction-free space for the interview to avoid windy noise. Turn off any notifications on your phone or computer to minimize disruptions.
5. Ensure that equipment is charged to avoid power issues.
6. Do not record interviews in any form. Any such act will be considered as a violation of the pledge you signed online and may invite punitive action from IIT Dharwad.

F.6. Research Areas for TA, FA, IS, EX and CT Categories

The applicant MUST indicate the choice of research areas as mentioned above in an order of preference. The candidates can apply to one or more categories (TA, FA, EX and CT) as per their eligibility.

F.6.1. Admission categories Available.

Admission Category	Advertised for Spring 26-27	Broad Research Area	Remarks
TA	Yes	All Research Areas mentioned in F.4.	-
PA	No	Not applicable	-
FA	Yes	All Research Areas mentioned in F.4.	-
IS	Yes	All Research Areas mentioned in F.4.	-
EX	Yes	All Research Areas mentioned in F.4.	-
CT	Yes	All Research Areas mentioned in F.4.	-

G. DEPARTMENT OF CHEMISTRY

G.1. Eligibility for Admission

G.1.1. Qualifying Degree

M.Sc. or equivalent degree in any area of chemical science/chemistry.
The candidates must also fulfill **any one** the following additional requirements:

- **Valid GATE Score or CSIR/UGC-NET-LS** or M.Phil or M.Tech./M.E. or equivalent degree in chemistry or related fields (for **TA category**)
- **Junior Research Fellowship** (JRF) of CSIR/ UGC/ DST INSPIRE/ DBT/ MHRD/ ICMR/ ICAR or any other relevant funding agencies (**for FA category**)

Positions are available in TA and FA categories. Note: The FA category vacancies are open to all social categories.

G.1.2. Minimum score in the qualifying degree

For General/EWS/OBC-NCL category candidates and/or for candidates where no concession in academic performance is called for, the eligibility criteria in the qualifying degree is First Class, as specified by the candidate's Institution/University. If the Institution/University does not specify the division/class, then one of the following will be considered as the eligibility criteria:

- a minimum of 60% marks (without round off) in aggregate. (OR)
- a minimum Cumulative Grade Point Average (CGPA) or Cumulative Performance Index (CPI) of 6.0 on the scale of 0-10; with corresponding proportional requirements when the scales are other than on 0-10, (for example, 4.8 on a scale of 0-8).

For SC/ST/PwD category candidates, a relaxation of 5% in the qualifying degree is applicable.

G.2. Modality of the Selection Process

Only the eligible applicants are permitted to participate in the selection process. One round of online interview will be conducted for the final selection after initial shortlisting of the candidates based on above criteria.

G.3. Syllabus

- Organic Chemistry
  Recommended textbooks: J. Clayden, N. Greeves, S. Warren
- Inorganic Chemistry
  Recommended textbooks: J. E. Huheey, E. A. Keiter, R. L. Keiter
- Physical Chemistry
  Recommended textbooks: Atkins' Physical Chemistry
- Spectroscopy
  Recommended textbooks: C. N. Banwell and D. L. Pavia

G.4. Focus area of research

The broad areas of research will include organic chemistry, biochemistry, and chemical biology, organometallic chemistry, bioinorganic chemistry, organic electronics, π-conjugated organic systems and computational chemistry. Students will have exposure to different interdisciplinary areas of chemistry, protein biochemistry and material science. The

Department of Chemistry admits Ph.D. candidates under the Teaching Assistantship (TA) and Fellowship Assistantship (FA) category for this round of admissions in the following research areas.

1. Organic Chemistry: CH activation of functional π-conjugated organic compounds, organometallics and organic synthetic chemistry, and bioorganic and medicinal chemistry are three currently active domains under this research area. Here, students will get exposure to different synthetic methodologies and various analytical techniques to characterize different organic molecules which have relevance in electronics, LEDs, catalysis, drug designing, and pharmaceutical industries.

Research supervisors: **PRof. Rajeswara Rao**, **Prof. Nilkamal Mahanta, Prof. Supriya Rej**

2. Physical Chemistry: The field of physical chemistry bridges experimental observations and theoretical predictions, aiding the development of new technologies across energy, materials, and biomedical sciences. Current research interests primarily focus on areas such as the design of organic electrodes for batteries and nitrogen reduction catalysts, computational material science, drug designing and discovery, artificial intelligence and machine learning, photochemistry, non-adiabatic dynamics, and polaritonic chemistry. To investigate these problems, a variety of computational tools are employed, including density functional theory (DFT), molecular dynamics simulations, quantum dynamics, QM/MM approaches, and multi-reference electronic structure methods. In addition, artificial intelligence and machine learning are used to accelerate drug discovery, predict chemical properties and reaction outcomes. In parallel, significant efforts are directed toward the development of new theoretical and computational methods to better describe chemical processes across these domains.

Research supervisors: **Prof. Sudhir Kumar Sahoo, Prof. Mahesh Gudem**, and **Prof. Anmol Kumar**

3. Inorganic Chemistry: Inorganic nanomaterials and new metal complexes have garnered significant attention for their diverse applications across various fields, including natural sciences, chemical industries, and biomedicine. Research teams of inorganic chemistry are currently engaged in developing metal complexes, atom precise metal nanomaterials and metal containing polymers for their application in catalysis, energy relevant small molecule activation, and sensing harmful gasses. Employing modern spectroscopic and experimental techniques, mechanistic pathways of the process are being investigated to generate new concepts. Researchers get exposure to learn various scientific skills: (i) expertise of inorganic, organometallics, nanocluster; (ii) multi-steps organic, metal-complex, and atom-precise metal nanocluster synthesis; (iii) use of Schlenk line and glove box to handle air and moisture sensitive reagents and reactions steps; (iv) isotopic labeling experiments etc.

Research supervisors: **PRof. Rajeswara Rao**, **Prof. Kundan K. Singh**, and **Prof. Supriya Rej**

4. Biological Chemistry: Enzymes are nature's chemists that carry out remarkable chemical reactions, particularly in the synthesis of antibiotics and other important compounds. Research groups are actively working on: (a) study of the organic chemistry of enzymes/proteins involved in the synthesis of biologically active compounds (anticancer, antibacterial, antifungal etc.) and enzyme mechanisms (such as C-H activation, C-C bond formation, molecular

rearrangement, amide bond activation, peptide bond formation etc.) involved in the synthesis of these natural products; Students will be exposed to different interdisciplinary principles and techniques of protein biochemistry and chemical biology in this field; (b) study of metalloenzymatic chemical processes (heme and non-heme oxygenases) using newly lab made structural and functional model complexes. This allows one to gain expertise of synthesis (sensitive multistep ligand and metal complex), hands-on experience of various spectroscopic characterization techniques, and application.

Research supervisors: **Prof. Nilkamal Mahanta**, and **Prof. Kundan K. Singh**.

G.5 Admission categories Available.

Admission Category	**Advertised for Spring 26-27**	**Broad Research Area**	**Remarks**
TA	Yes	All Research Areas mentioned in G.4.	-
PA	No	Not applicable	-
FA	Yes	All Research Areas mentioned in G.4.	-
IS	No	Not applicable	-
EX	No	Not applicable	-
CT	No	Not applicable	-

H. DEPARTMENT OF CIVIL AND INFRASTRUCTURE ENGINEERING (CIE)

H.1. Eligibility Criteria

H.1.1. Qualifying Degree

M.Tech./M.E. or equivalent degree in civil engineering or civil and infrastructure engineering or civil and environmental engineering or any related stream.

H.1.2. Minimum Score in Qualifying Degree and GATE Qualification

For General/EWS/OBC-NCL category candidates and/or for candidates where no concession in academic performance is called for, the eligibility criteria in the qualifying degree (M.Tech. /M.E.):

- A minimum of 60% marks (without rounding off) in aggregate, OR,
- A minimum Cumulative Grade Point Average (CGPA) or Cumulative Performance Index (CPI) of 6.0 on the scale of 0-10, with corresponding proportional requirements when the scales are other than on 0-10 (for example, 4.8 on a scale of 0-8).
- Candidates must qualify in GATE at least once (Candidate need not to have a valid GATE score) OR Candidate must have a CGPA $\geq$ 7.5 (or equivalent percentage) in master's OR Candidate must have one SCI-indexed journal publication.

For SC/ST category candidates and differently abled candidates (PwD), a relaxation of 5 % (or CPI/CGPA of 0.5 on the scale of 0-10) in the qualifying degree is applicable. However, candidates must qualify in GATE at least once (Candidate need not to have a valid GATE score) OR Candidate must have a CGPA $\geq$ 7.5 (or equivalent percentage) in master's OR Candidate must have one SCI-indexed journal publication.

H.2. Eligibility of applicants in the final phase of getting the qualifying degree

Students who are in the final phase of receiving the above-mentioned qualifying degree and who are likely to graduate before the commencement of the Academic Semester (2025-2026) of IIT Dharwad are also eligible to apply. However, if offered, the admission to those candidates would be provisional. To join an academic program at IIT Dharwad, such candidates need to furnish the necessary documents regarding completion of the degree on the date of joining mentioned in Section A above. They need to meet the criteria specified in the section above, considering the updated score in the qualifying degree. In the meanwhile, the aggregate academic performance announced by the respective university till the last date for submission, should be used to determine eligibility for application and same to be reported in the online application.

H.3. Modality of selection process

Only eligible applicants are permitted to participate in the selection process. The shortlisted candidates will be called for an interview by the respective panel based on the research area preference mentioned in the admission form. The selection process would involve two rounds (online mode).

Round 1: Candidates must present their own research work for 10 minutes. Instructions: (a) The

First slide must contain candidates' brief biodata. (b) The last slide must contain a prospective Ph.D. research problem statement.

Round 2: The selected candidate from Round 1 will be called for a technical interview. The technical interview will be based on the GATE syllabus (Civil Engineering or Equivalent Streams).

The candidates are encouraged to check the Institute's Website from time to time. Selection committee decisions are final in all matters, including any disciplinary matters/malpractice.

H.4. Research Areas for TA, IS, EX and CT Categories

The research areas are broadly classified into three streams as described below. The applicant MUST indicate their research area choices in order of preference.

1. Structural Engineering and Materials
2. Transportation Engineering
3. Geotechnical Engineering

Note:

- The candidates can apply to one or more categories (TA, PA, EX, and CT) based on their eligibility.

H.5. Admission Categories Available

Admission Category	**Advertised for Spring 26-27**	**Broad Research Area**	**Remarks**
TA	Yes	All Research Areas mentioned in H.4.	-
PA	No	Not applicable	-
FA	No	Not applicable	-
IS	Yes	All Research Areas mentioned in H.4.	-
EX	Yes	All Research Areas mentioned in H.4.	-
CT	Yes	All Research Areas mentioned in H.4.	-

I. DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (CSE)

I.1. Qualifying Degree

M.Tech. or equivalent degree in Computer Science and Engineering or any related stream*. (or)

M.Sc. in Computer Science/Mathematics/Statistics or equivalent with a **valid GATE score or valid UGC/CSIR NET JRF/LS scorecard**

I.1.1 Minimum score in the qualifying degree

For General/EWS/OBC-NCL category candidates and/or for candidates where no concession in academic performance is called for, the eligibility criteria in the qualifying degree (M.Tech./M.E.):

- A minimum of 60% marks (without round off) in aggregate, OR,
- A minimum Cumulative Grade Point Average (CGPA) or Cumulative Performance Index (CPI) of 6.0 on the scale of 0-10; with corresponding proportional requirements when the scales are other than on 0-10, (for example, 4.8 on a scale of 0-8).

For SC/ST category candidates and differently abled candidates (PwD), a relaxation of 5% (or CPI/CGPA of 0.5 on the scale of 0-10) in the qualifying degree is applicable.

***The definition of related stream is at the discretion of the Department of Computer Science and Engineering, IIT Dharwad.**

I.1.2. Eligibility of applicants in the final phase of getting the qualifying degree

Students who are in the final phase of receiving the above-mentioned qualifying degree and who are likely to graduate before commencement of Spring 2026-27 semester of IIT Dharwad are also eligible to apply. However, if offered, admission to those candidates would be provisional. To join an academic program at IIT Dharwad, such candidates need to furnish necessary documents regarding completion of the degree on the date of joining mentioned in Section A above. They need to meet the criteria specified in the section above considering updated score in the qualifying degree. In the meanwhile, the aggregate academic performance announced by the respective university till the last date for submission mentioned in section A should be used to determine eligibility for application and same to be reported in the online application.

I.2. Modality of selection process

Only eligible applicants are permitted to participate in the selection process. The selection process would involve two rounds; round-1: An online interview to test the aptitude, programming skills and knowledge of discrete structures, data structures and algorithms of the candidate; round-2: The shortlisted candidates from round-1 will be called for interview (online) by the respective panel based on the research area preference mentioned in the admission form. The candidates are encouraged to check the Institute website https://www.iitdh.ac.in/doctoral-0 from time to time. Selection committee decisions are final in all matters including any disciplinary matters/malpractice.

I.3. Focus area of research

The research topics are broadly classified as given below. The applicant may be asked to

indicate the choice of research topics in order of preference.

1. Computer Networks, IoT and Security (CN):

- 5G/IoT Networks, AI Driven Networking, Network Virtualization, Network/Cyber Security, Blockchains, Software Defined Networks, Network Function Virtualization, Data Center Networking.

2. Computer Systems and Programming:

- Approximate Computing, Edge-Cloud systems, Modeling and characterization of heterogeneous processors, Runtime Verification of Hardware and Efficient Computer Architectures, Parallel Computing, Compilers and Translation Systems, Programming models and runtime systems.

3. Machine learning (ML) and Artificial Intelligence (AI):

- Machine Learning (ML), Deep Learning (DL), Reinforcement Learning (RL), Computer Vision, Stochastic Control and Optimization, Bayesian Optimization, Text Mining, Speech and Audio Processing.
- Handwriting and Document Processing, Natural Language Processing, Machine Translation, LLMs, AI Chatbot Assistant, Computer Vision, ML for Cyber Physical Systems, AR/VR/MX, Mining large data streams, ML for Cyber Security, Big Data Analytics, Distributed data processing.

4. Systems for Machine Learning (SysML)

- Application of neural networks on Edge devices, Hardware for machine learning systems; GPU/TPU/NPU/ML systems and software stack, quantized and low precision machine learning.

5. Theoretical Computer Science (TCS):

- Algorithms, Concurrency, Formal Verification, Verification of quantum programs, Graph Theory, Logic.

I.4. Admission Categories Available

Admission Category	Advertised for Spring 26-27	Broad Research Area	Remarks
TA	Yes	All Research Areas mentioned in **I.3.**	The applicant may be asked to indicate the choice of research topics in the order of preference.
PA	No	Not Applicable	-
FA	No	Not Applicable	-
IS	No	Not Applicable	-
EX	Yes	All Research Areas mentioned in **I.3.**	-
CT	No	Not Applicable	-

I.5. Syllabus

- **Discrete Mathematics:** Propositional and first order logic. Sets, relations, functions, partial orders, and lattices. Groups. Graphs: connectivity, matching, coloring. Combinatorics: counting, recurrence relations, generating functions, Linear Algebra: Matrices, determinants, system of linear equations, eigenvalues and Eigenvectors, LU decomposition. Calculus:

Limits, continuity, and differentiability. Maxima and minima. Mean value theorem. Integration. Probability: Random variables. Uniform, normal, exponential, Poisson, and binomial distributions. Mean, median, mode and standard deviation. Conditional probability and Bayes theorem.

- **Computer Organization and Architecture:** Machine instructions and addressing modes. ALU, data-path and control unit. Instruction pipelining. Memory hierarchy: cache, main memory, and secondary storage; I/O interface (interrupt and DMA mode).
- **Programming and Data Structures:** Programming in C. Recursion. Arrays, stacks, queues, linked lists, trees, binary search trees, binary heaps, graphs. In
- **Algorithms:** Searching, sorting, hashing. Asymptotic worst-case time and space complexity. Algorithm design techniques: greedy, dynamic programming and divide-and-conquer. Graph search, minimum spanning trees, shortest paths.
- **Theory of Computation:** Regular expressions and finite automata. Context-free grammar and push-down automata. Regular and context-free languages, pumping lemma. Turing machines and undecidability.
- **Compiler Design:** Lexical analysis, parsing, syntax-directed translation. Runtime environments. Intermediate code generation.
- **Operating System:** Processes, threads, inter-process communication, concurrency and synchronization. Deadlock. CPU scheduling. Memory management and virtual memory. File systems.
- **Computer Networks:** Concept of layering. LAN technologies (Ethernet). Flow and error control techniques, switching. IPv4/IPv6, routers and routing algorithms (distance vector, link state). TCP/UDP and sockets, congestion control. Application layer protocols (DNS, SMTP, POP, FTP, HTTP). Basics of Wi-Fi. Network security: authentication, basics of public key and private key cryptography, digital signatures and certificates, firewalls.

J. DEPARTMENT OF ELECTRICAL, ELECTRONICS AND COMMUNICATION ENGINEERING (EECE)

J.1. Eligibility Criterion

J.1.1. Qualifying Degree

M.Tech., M.S., M.E. or equivalent degree in Electrical Engineering, Electronics and Communication Engineering, Electrical and Electronics Engineering, Instrumentation Engineering, Computer Science and Engineering, or any related stream.
OR
M.Sc. in Mathematics / Statistics / Electronics or equivalent with valid GATE score or valid **UGC/CSIR NET JRF/LS** scorecard.

J.1.2. Minimum score in the qualifying degree

For General/EWS/OBC-NCL category candidates and/or for candidates where no concession in academic performance is called for, the eligibility criteria in the qualifying degree (M.Tech./M.E/MSc) is as follows:

- a minimum of 60% marks (without round off) in aggregate, OR,
- a minimum Cumulative Grade Point Average (CGPA) or Cumulative Performance Index (CPI) of 6.0 on the scale of 0-10; with corresponding proportional requirements when the scales are other than on 0-10, (for example, 4.8 on a scale of 0-8).

For SC/ST category candidates and differently abled candidates (PwD), a relaxation of 5% (or CPI/CGPA of 0.5 on the scale of 0-10) in the qualifying degree is applicable.

J.1.3. Eligibility for Admission - Direct Ph.D.

Students can apply for the Ph.D. program directly after completing their B.Tech./B.E degrees for a direct Ph.D. program. The following are the eligibility criteria for the same:

A. B.Tech. in one of the following streams:

- Electrical Engineering,
- Electronics and communication Engineering,
- Instrumentation engineering
- Instrumentation and control engineering
- Tele-communication engineering
- Electronics and Telecommunication engineering
- Electrical and Electronics
- Computer Science
- Data Science and Artificial Intelligence

B. A valid GATE score in one of the following papers

- Electrical Engineering (GATE Paper code: EE)
- Electronics and Communication Engineering (GATE Paper code: EC)
- Instrumentation Engineering (GATE Paper code: IN)
- Data Science and Artificial Intelligence (GATE Paper code: DA)
- Computer Science & Information Technology (GATE Paper code: CS)

Valid GATE score is essential for all the candidates, except for candidates who have B.Tech. or equivalent degree from IITs or BS degree from IISc with a minimum CPI/CGPA of 8.0 on the scale of 10.

J.1.4. Eligibility of applicants who are in the final phase of getting the qualifying degree

Students who are in the final phase of receiving the above-mentioned qualifying degree and who are likely to graduate before commencement of Spring 2026-27 semester of IIT Dharwad are also eligible to apply. However, if offered, the admission to those candidates would be provisional. To join an academic program at IIT Dharwad, such candidates need to furnish necessary documents regarding completion of the degree on the date of joining at IIT Dharwad. They need to meet the criteria specified in section above considering an updated score in the qualifying degree. In the meanwhile, the aggregate academic performance announced by the respective university till the last date for submission should be used to determine eligibility for application and same to be reported in the online application.

J.1.5. Application Categories and Financial Support

The Department of Electrical, Electronics and Communication Engineering offers admission to Ph.D. programs under TA, PA, FA, EX, CT, Institute Staff, and Visvesvaraya PhD Fellowship categories. The department will try to accommodate the candidates' order of preferences for these categories, but does not guarantee the same. The details of each application category are given in Section APPLICATION CATEGORIES & FINANCIAL SUPPORT of this document.

Note: **The PA, EX, FA, CT, IS, and Visvesvaraya PhD Fellowship category vacancies are open to all social categories.**

J.1.6. Admission Categories Available

Admission Category	Advertised for Spring 26-27	Broad Research Area	Remarks
TA	Yes	Research Areas mentioned in **J.3.1.**	-
PA	Yes	Research area mentioned in **J.3.2**	Vacancies: 02
FA	Yes	Research Areas mentioned in **J.3.1.**	-
IS	Yes	Research Areas mentioned in **J.3.1.**	-
EX	Yes	Research Areas mentioned in **J.3.1.**	-
CT	Yes	Research Areas mentioned in **J.3.1.**	-
Visvesvaraya PhD Fellowship	Yes	Research Areas mentioned in **J.3.1.** under Microelectronics and VLSI	Vacancy: 01

J.2. Guidelines for Shortlisted Candidates

J.2.1. Selection Process

All the eligible candidates will be invited for the interview via video conferencing (online-mode). Depending on the number of applications received, there may be additional rounds of interviews (online-mode). Syllabus for the interview is given in Section K.4 of this document.

The interview slot (date and starting time) specific to each candidate will be communicated online at https://www.iitdh.ac.in/Ph.D..
The selection committee decision is final in all matters including any disciplinary matters/malpractice.

J.3. Research Areas

J.3.1. For TA, FA, EX, CT and IS categories

The research areas are broadly classified in streams as described below. **The applicant MUST indicate the choice of research topics in an order of preference**.

1. Microelectronics and VLSI

- ***VLSI Circuits:*** Including but not limited to, Analog / Mixed signal / RF Integrated Circuits and Systems, Power management and Energy harvesting circuits, Digital signal processing for digital VLSI, and AI/ML hardware accelerators design, etc.
- ***Electronic Devices:*** Including but not limited to Gas sensors, Nano-electronics, Flexible devices, GaN-based High-electron mobility transistors (HEMTs), Silicon Carbide (SiC) Power Diodes, Semiconductor Radiation Detectors etc.

2. Communication, Signal Processing, and Machine Learning

- ***Communication Technologies:*** Including but not limited to, physical and medium access control layer technologies in next generation wireless systems, internet of things, multiple access methods, multi-input multi-output systems, millimeter wave communications, low-latency and energy harvesting based communications, machine learning and blockchain oriented resource allocation in 6G, quantum communication.

- ***Signal Processing and Machine Learning:*** Including but not limited to, speech and natural language processing, image processing and computer vision, and machine learning and deep learning for signal processing.

3. Power Electronics, Energy, and Power Systems

- ***Power systems:*** Grid-connected renewables and flexible loads, grid-forming technology, autonomous microgrids and smart grids, power system stability, control, and protection.

- ***Power Electronics and Drives:*** Converters for grid-interfacing, modular and multi-level inverters, power converters for electric vehicles, DC circuit breakers for medium-voltage applications, design of wide-bandgap device-based converters, modeling, and controls for advanced power electronics.

- ***Magnetics:*** Design of electric machines, magnetics for power-electronics, multiphysics optimization, eddy-current devices, Condition monitoring, characterization of magnetic components, rotational and stress-induced losses.

4. Control and Robotics

- Including but not limited to Control of Robots through Speech Signals, Autonomous Vehicles, Control for Differential Games, Control of Structures etc.

J.3.2. For PA category*

Applications are called in following broad domains:

1. Microelectronics and VLSI – 1 position
2. Communication, Signal processing and Machine learning – 1 position
3. Power Electronics, Energy, and Power Systems – 0 position
4. Control and Robotics – 0 position

***The number of seats under the PA category may change depending on the approval of projects and availability of funds.**

J.4. Interview Syllabus

All applicants should choose one stream for the interview while submitting the online application form.

Common for all the streams

1. General aptitude, reasoning, and comprehension
2. **Engineering Mathematics:** Matrix Algebra, Systems of linear equations, Eigenvalues, Eigenvectors, Concepts from integration and differentiation, Fourier Transform and Laplace Transform.

Stream 1: Communication and Signal Processing

1. Basic Electrical Networks:

KCL, KVL, Node and Mesh analysis, Network theorems etc.

2. Signals and Systems:

Continuous-time signals: Fourier series and Fourier transform representations, sampling theorem and applications.
Discrete-time signals: discrete-time Fourier transform (DTFT), DFT, FFT, z- transform and sampling theorem.
LTI systems: definition and properties, causality, stability, impulse response, convolution, poles and zeros and frequency response.
Random processes: basics of probability, random variables, CDF, PDF, random processes, mathematical expectation, conditional probability, and conditional expectation.

3. Communication:

Random processes: Basics of probability, random variables, CDF, PDF, random processes, mathematical expectation, conditional probability, and conditional expectation.
Digital communications: Digital modulation schemes, MAP and ML decoding, notions of bandwidth, SNR and BER for digital modulation, fundamentals of error correction codes (e.g.: Linear Block Codes like Hamming code).

Stream 2: Electronic Devices and Mixed signal ASIC Design

1. Basic Electrical Networks:

KCL, KVL, Node and Mesh analysis, Network theorems etc.

2. Electronic Devices:

Energy bands in intrinsic and extrinsic silicon; Carrier transport: diffusion current, drift current, mobility, and resistivity; Generation and recombination of carriers; Poisson and continuity equations; P-N junction, Zener diode, BJT, MOS capacitor, MOSFET, LED, photo diode and solar cell; Integrated circuit fabrication process: oxidation, diffusion, ion implantation, photolithography and twin-tub CMOS process.

3. Analog Circuits:
Basics of Analog circuits.

4. Digital Systems:
Number systems; Combinatorial circuits; Sequential circuits.

Stream 3: Power and Energy Systems

1. Electric Circuits:
KCL, KVL, Node and Mesh analysis, Transient response of dc and ac networks, Sinusoidal steady-state analysis, Resonance, Ideal current and voltage sources, Thevenin's theorem, Norton's theorem, Superposition theorem, Maximum power transfer theorem, three phase circuits, Power and power factor in ac circuits.

2. Power Electronics:
Characteristics of MOSFET, IGBT and diode, DC to DC conversion: Buck, Boost and Buck-Boost converters; Single and three phase configuration of uncontrolled rectifiers, Line commutated thyristor-based converters.

3. Power Systems:
Per-unit quantities, Newton-Raphson load flow methods, Voltage and Frequency control, Power factor correction, Symmetrical components and fault analysis, Power System Stability, Power System Protection.

4. Electrical Machines:
Single phase transformer: equivalent circuit, phasor diagram, open circuit and short circuit tests, regulation and efficiency; Three phase transformers: connections, parallel operation; Three phase induction motors: principle of operation, types, performance, torque-speed characteristics, no-load and blocked rotor tests, equivalent circuit, starting and speed control; Synchronous machines: cylindrical and salient pole machines, performance, regulation, starting of synchronous motor, characteristics, P&Q Control.

K. DEPARTMENT OF MECHANICAL, MATERIALS AND AEROSPACE ENGINEERING (MMAE)

The Department of Mechanical, Materials and Aerospace Engineering (MMAE) invites applications for its Ph.D. program. While the Regular Ph.D. program is tailored for candidates holding a Master's degree, the Department also offers a Direct Ph.D. program for candidates applying directly with an undergraduate degree without needing a Master's degree.

K.1. Regular Ph.D.

K.1.1. Eligibility criteria

M.Tech./M.E./M.Sc. (Engg.) or equivalent degree in Mechanical, Materials, Metallurgical, Aerospace, Civil, Biomedical, Electrical, Computer Science, Electronics, Chemical Engineering, or equivalent stream.

K.1.2. Qualifying criteria

The score requirement in the qualifying degree (M.Tech./M.E./M.Sc. (Engg.) or equivalent) is as follows:

- A minimum of 60% marks (without rounding off) in aggregate, OR,
- A minimum Cumulative Grade Point Average (CGPA) or Cumulative Performance Index (CPI) of 6.0 on the scale of 0-10; with corresponding proportional requirements when the scales are other than on 0-10, (for example, 4.8 on a scale of 0-8).

For SC/ST category candidates and differently abled candidates (PwD), a relaxation of 5% (or CPI/CGPA of 0.5 on the scale of 0-10) in the qualifying degree is applicable.

K.2. Direct Ph.D.

K.2.1. Eligibility criteria

Students can apply for the Direct Ph.D. program upon completion of their B.Tech./B.E degrees for a Direct Ph.D. program with a valid GATE score.

K.2.2. Qualifying criteria

- B.Tech./B.E. or equivalent degree in Mechanical Engineering, Materials and Metallurgical Engineering, Aerospace Engineering, or other related streams with the following score requirements [#]:
    - Marks based score: a minimum of 60% marks (without roundoff) in aggregate, or
    - Cumulative Grade Point Average (CGPA) or Cumulative Performance Index (CPI) based score: a minimum of 6.0 on a scale of 0 to 10 or proportional requirement if the scale is other than 0 to 10 (for example, 4.8 on a scale of 0-8).
- A valid GATE score in one of the following papers AE, ME, MT, PI, XE (XE-A plus any two from XE-B, XE-C, XE-D and XE-E) [*]

[#] **For SC/ST category candidates and differently abled candidates (PwD), a relaxation of 5% (or CPI/CGPA of 0.5 on the scale of 0-10) in the qualifying degree is applicable.**

[*] A valid GATE score is essential for candidates applying in the TA or PA category. However, GATE score can be waived for B.E./B.Tech. graduates of CFTI (Centrally Funded Technical Institute), having CPI/CGPA 8 (on scale of 10) or above.

Note:Candidates who are in their final year of their qualifying degree may refer to the section "Eligibility of applicants in the final phase of getting the qualifying degree" to know about their eligibility.

K.3. Application categories

The Department of MMAE offers admission to Ph.D. programs under TA, FA, EX, CT, SP, IS, and Visvesvaraya PhD Fellowship categories. The Department will try to accommodate the candidates' order of preferences for these categories, but cannot guarantee it. The details of each application category are given in Section **B.** APPLICATION CATEGORIES & FINANCIAL SUPPORT of this document.
Note: The EX, FA, CT, IS, and Visvesvaraya Ph.D. Fellowship category vacancies are open to all social categories.

K.3.1. Admission Categories Available

Admission Category	Advertised for Spring 26-27	Broad Research Area	Remarks
TA	Yes	Research Areas mentioned in **K.4**	-
PA	No	Not Available	Vacancies: 0
FA	Yes	Research Areas mentioned in **K.4**	-
IS	Yes	Research Areas mentioned in **K.4**	-
EX	Yes	Research Areas mentioned in **K.4**	-
CT	Yes	Research Areas mentioned in **K.4**	-
Visvesvaraya Ph.D. Fellowship	Yes	Research Areas mentioned in **K.4**	Vacancy: 1

K.4. Areas of research

The following streams are available for candidates applying to PhD. Applicants should be interested in at least one of the following research areas, listed below:

1. Design
2. Thermal and Fluids
3. Manufacturing and Materials
4. Computational Fluid Dynamics
5. Computational Materials Science
6. Robotics and Intelligent Systems

The tentative list of research areas and the associated faculty in each of the above streams is available under "Research Streams" at the Department Webpage: https://mmae.iitdh.ac.in/phd.html.

K.5. Eligibility of applicants in the final phase of getting the qualifying degree

Students who are in the final phase of receiving the above-mentioned qualifying degree and who are likely to graduate before commencement of Spring 2026-27 semester of IIT

Dharwad are also eligible to apply. However, if offered, admission to those candidates would be provisional. To join an academic program at IIT Dharwad, such candidates need to furnish necessary documents regarding completion of the degree on the date of joining mentioned in Section A above. They need to meet the criteria specified in the section above considering an updated score in the qualifying degree. In the meanwhile, the aggregate academic performance announced by the respective university till the last date for submission mentioned in section A should be used to determine eligibility for application and same to be reported in the online application.

K.6. Modality of selection process

The list of shortlisted candidates will be available on the Institute website and all such candidates will also be communicated through email address provided in the application form.. Only candidates shortlisted after the first round will be called for the second round. The online meeting link and the time schedule will be communicated to the candidates through the email.

Candidates must attend the interviews on a desktop or laptop computer with a webcam, a speaker, and a microphone compatible with Google Meet. Candidates are not permitted to refer to books, online material, or AI-based tools, nor to communicate with any other person, throughout the interview. The candidates may be remotely proctored via the webcam and screen-sharing options. Any suspicious activity indicating cheating during the first or second rounds of selection will be grounds for disqualification of candidature.

First round of interview: The candidates will be asked to give a short presentation on their area(s) of expertise (such as B.Tech./M.Tech. projects or any previous work experience), followed by a technical interaction.

Second round of interview: Candidates will be asked technical questions based on the syllabus specified for the respective stream and/or area of interest, in addition to the Common Syllabus.

K.7. DOs & DON'Ts for Your Online PhD Interview

DOs:

- **Test your setup beforehand:** Have a mock call to ensure your audio, video, and internet connection (at least 2 Mbps) are working smoothly. Keep at least 2 GB of data handy if not on Wi-Fi.
- **Prepare your space:** Sit in a quiet, well-lit room with a neutral background (a wall or virtual background works). Ensure your camera is at eye level.
- **Keep essentials close:** Have your pen, paper, non-programmable scientific calculator, and a glass of water within reach. Ensure your device is fully charged or plugged in.
- **Join early and stay focused:** Log in to Google Meet 10-15 minutes before the start of the session. Keep your video ON during the interview and stay muted when not speaking.
- **Remember:** The use of AI tools, unauthorized additional devices, or any unauthorized assistance during the interview is strictly prohibited and will be treated as a violation of academic integrity.

DON'Ts:

- **Do not cheat:** The use of AI tools (like ChatGPT or other language models), smart devices, or seeking help from anyone during the interview is strictly forbidden. Any

such act will be considered a serious violation and will lead to disqualification and possible blacklisting.

- **Do not record:** Recording the interview in any form is strictly prohibited and will invite punitive action.
- **Do not disappear:** Stay visible on camera at all times. Do not leave your seat or take any breaks during the interview.
- **Do not have others nearby:** Ensure you are alone in the room. Any interaction with someone else during the interview will be considered suspicious.
- **Do not ask about results or feedback:** The results of a round will be announced online as soon as possible. Use your time with the panel for meaningful academic inquiries instead.

Note: For any matter related to the selection process, the decision of the selection committee is final.

K.8. Syllabus: Common syllabus for all streams

K.8.1. Engineering Mathematics

Linear Algebra: Matrix algebra, systems of linear equations, eigenvalues and eigenvectors.

Calculus: Functions of single variable, limit, continuity and differentiability, mean value theorems, indeterminate forms; evaluation of definite and improper integrals; double and triple integrals; partial derivatives, total derivative, Taylor series (in one and two variables), maxima and minima, Fourier series; gradient, divergence and curl, vector identities, directional derivatives, line, surface and volume integrals, applications of Gauss, Stokes and Green's theorems.

Differential equations: First Order Equations (linear and nonlinear); higher order linear differential equations with constant coefficients; Euler-Cauchy equation; initial and boundary value problems; Laplace transforms; solutions of heat, wave and Laplace's equations.

Complex variables: Analytic functions; Cauchy-Riemann equations; Cauchy's integral theorem and integral formula; Taylor and Laurent series.

Probability and Statistics: Definitions of probability, sampling theorems, conditional probability; mean, median, mode and standard deviation; random variables, binomial, Poisson and normal distributions.

Numerical Methods: Numerical solutions of linear and non-linear algebraic equations; integration by trapezoidal and Simpson's rules; single and multi-step methods for differential equations.

K.8.2. Analytical reasoning

Verbal reasoning: reading comprehension, drawing inferences based on multiple facts stated in short paragraphs.

Non-verbal reasoning: inductive, logical, abstract, diagrammatic, and spatial reasoning.

K.9. Syllabus: Stream-specific syllabus

K.9.1. Design stream

Engineering Graphics: Orthographic projections of lines, planes and solids, true length and true angle, sections of solids and intersections of solids, solid modeling.

Engineering Mechanics: Free-body diagrams and equilibrium; trusses and frames; virtual work; kinematics and dynamics of particles and of rigid bodies in plane motion; impulse and momentum (linear and angular) and energy formulations, collisions.

Mechanics of Materials: Stress and strain, elastic constants, Poisson's ratio; Mohr's circle for plane stress and plane strain; thin cylinders; shear force and bending moment diagrams; bending and shear stresses; deflection of beams; torsion of circular shafts; Euler's theory of columns; energy methods; thermal stresses; strain gauges and rosettes; testing of materials with universal testing machine; testing of hardness and impact strength.

Theory of Machines: Displacement, velocity, and acceleration analysis of plane mechanisms; dynamic analysis of linkages; cams; gears and gear trains; flywheels and governors; balancing of reciprocating and rotating masses; gyroscope. Vibrations: Free and forced vibration of single degree of freedom systems, effect of damping; vibration isolation; resonance; critical speeds of shafts.

Machine Design: Design for static and dynamic loading; failure theories; fatigue strength and the S-N diagram; principles of the design of machine elements such as bolted, riveted, and welded joints; shafts, gears, rolling and sliding contact bearings, brakes and clutches, springs.

K.9.2. Thermal and Fluids stream

Fluid Mechanics: Fluid properties; fluid statics, manometry, buoyancy, forces on submerged bodies, stability of floating bodies; control-volume analysis of mass, momentum and energy; fluid acceleration; differential equations of continuity and momentum; Bernoulli's equation; dimensional analysis; viscous flow of incompressible fluids, boundary layer, elementary turbulent flow, flow through pipes, head losses in pipes and bends, flow in convergent-divergent channels, vorticity and stream-functions, elementary Computational Fluid Dynamics, finite-difference approximation to the first and second order partial derivatives.

Heat transfer: Modes of heat transfer; one dimensional heat conduction, resistance concept and electrical analogy, heat transfer through fins; unsteady heat conduction, lumped parameter system, Heisler's charts; thermal boundary layer, dimensionless parameters in free and forced convective heat transfer, heat transfer correlations for flow over flat plates and through pipes, effect of turbulence; heat exchanger performance, LMTD and NTU methods; radiative heat transfer, Stefan- Boltzmann law, Wien's displacement law, black and gray surfaces, view factors radiation network analysis.

Thermodynamics: Thermodynamic systems and processes; properties of pure substances, behavior of ideal and real gasses; zeroth and first laws of thermodynamics, calculation of work and heat in various processes; second law of thermodynamics; thermodynamic property charts and tables, availability, and irreversibility; thermodynamic relations.

Applications Power Engineering: Air and gas compressors; vapor and gas power cycles, concepts of regeneration and reheat. I.C. Engines: Air-standard Otto, Diesel, and dual cycles.

Refrigeration and air-conditioning: Vapour and gas refrigeration and heat pump cycles; properties of moist air, psychrometric chart, basic psychrometric processes.

Turbomachinery: Impulse and reaction principles, velocity diagrams, Pelton-wheel, Francis and Kaplan turbines, Steam and gas turbines, axial flow compressors.

K.9.3. Manufacturing and Materials stream

Engineering Materials: Structure and properties of engineering materials, Crystal Imperfections, phase diagrams, heat treatment, stress-strain diagrams for engineering materials. Dislocation theory, Strengthening mechanisms, fracture mechanics, fractography, ductile to brittle transition. Fatigue, Mechanisms of high temperature deformation and failure, X-ray Diffraction.

Metal Forming: Plastic deformation and yield criteria; fundamentals of hot and cold working processes; load estimation for bulk (forging, rolling, extrusion, drawing) and sheet (shearing, deep drawing, bending) metal forming processes, Plastic deformation by slip and twinning.

Sheet Metal working: Die and punch clearances, blanking, piercing, punching, bending, cup drawing, coining, embossing, incremental forming.

Metal Casting: Different types of casting, solidification and cooling, Pattern materials, allowances, types of patterns, cores, element of gating systems, types of gates, riser design considerations, casting defects.

Polymers and Composites: Thermoplastics, thermosets, elastomers and composites, gradient material, and related processes.

Computer Integrated Manufacturing: Basic concepts of CAD/CAM and their integration tools, tool path generation, additive manufacturing.

K.9.4. Computational Fluid Dynamics stream

Basics of Fluid Mechanics, Basics of Heat and Mass Transfer, Navier-Stokes equations, Mathematical modeling, Partial Differential Equations (PDEs), Algebraic solution of PDEs, finite-difference method, Finite-Volume method, Basic C, C++, Matlab, or Python programming.

K.9.5. Computational Materials Science stream

Deformation, Elastic and Plastic deformation, Stress and Strain tensors, Finite element analysis, Constitutive modeling, Multiscale modeling, Defect dynamics, Dislocation mechanics and dynamics, Mechanical Equilibrium, Mechanics of Materials, Homogenization, Differential and Integral Calculus, Differential Operators, Scientific Programming language, Linear System of Equations, Matrix Algebra, Numerical methods, Fracture analysis, Density functional theory, Structure-property relationships, Cohesive zone modeling, All-atom simulations, Structure and properties of engineering materials, Crystal Imperfections, phase diagrams, heat treatment, stress-strain diagrams for engineering materials, Strengthening mechanisms, fracture mechanics, fractography, ductile to brittle transition, Mechanisms of

high temperature deformation and failure, X-ray Diffraction, failure theories, fatigue strength and the S-N diagram, testing of materials.

K.9.6. Robotics and Intelligent Systems stream

Engineering Graphics: Orthographic projections of lines, planes and solids, true length and true angle, sections of solids and intersections of solids, solid modeling.

Engineering Mechanics: Free-body diagrams and equilibrium; trusses and frames; virtual work; kinematics and dynamics of particles and of rigid bodies in plane motion; impulse and momentum (linear and angular) and energy formulations, collisions.

Mechanics of Materials: Stress and strain, elastic constants, Poisson's ratio; Mohr's circle for plane stress and plane strain; thin cylinders; shear force and bending moment diagrams; bending and shear stresses; deflection of beams; torsion of circular shafts; Euler's theory of columns; strain gauges and rosettes; testing of hardness and impact strength.

Theory of Machines: Displacement, velocity, and acceleration analysis of plane mechanisms; dynamic analysis of linkages; cams; gears and gear trains; Vibrations: Free and forced vibration of single degree of freedom systems, effect of damping; vibration isolation; resonance.

Machine Design: Design for static and dynamic loading; failure theories; fatigue strength and the S-N diagram; principles of the design of machine elements such as bolted, riveted and welded joints; shafts, gears, springs.

Control System: S-plane diagrams, root locus plots, PID controls, 1st and 2nd order system response.

Data analysis and synthesis: Basic coding skills, introduction to data structure and algorithms, probability and statistics.

L. DEPARTMENT OF PHYSICS

L.1. Eligibility for Admission

L.1.1. Qualifying Degree

- M.Sc. or equivalent degree in Physics / Applied Physics / Photonics / Solid State Physics / or other topics in Physics.
- M.Tech. / M.S. in Optics / Optoelectronics / Photonics / Engineering Physics / Electrical Engineering/ Remote Sensing & GIS/ Geoinformatics or other topics in applied Physics

The candidates who do not have M.Tech. degree must also fulfill ONE of the following additional requirements:

➔ **For TA Category**: Valid GATE Score **OR** valid CSIR/UGC- National Eligibility Test (NET) / Lectureship (LS) **OR** valid Joint Entrance Screening Test (JEST) score

➔ **For FA Category**: Junior Research Fellowship (**JRF**) or equivalent fellowship of CSIR / UGC / DST INSPIRE or any other funding agencies in Physical Sciences.

L.1.2. Minimum score in the qualifying degree

For General/EWS/OBC-NCL category candidates and / or for candidates where no concession in academic performance is called for, the eligibility criteria in the qualifying degree is First Class, as specified by the candidate's Institution/University. If the Institution/University does not specify the division/class, then one of the following will be considered as the eligibility criteria:

- A minimum of 60% marks (without round-off) in aggregate. (OR)
- A minimum Cumulative Grade Point Average (CGPA) or Cumulative Performance Index (CPI) of 6.0 on a scale of 0-10, with corresponding proportional requirements when the scales are other than 0-10 (for example, 4.8 on a scale of 0-8).
- A 5% relaxation in the qualifying degree is applicable for SC/ST/PwD category candidates.

L.2. Modality of Selection Process

The Department of Physics admits Ph.D. candidates under the full-time research scholarship - Teaching Assistantship (TA), Fellowship Awardee (FA), and Externally Sponsored (EX).

L.2.1. Guidelines for shortlisted candidates

For all the categories, based on the information provided by the applicants, a short-list of candidates for the selection process will be prepared. The list will be declared on the Institute website on the date specified in the schedule. Only the short-listed candidates are permitted to participate in the selection process.

L.2.2. Admission Categories Available

Admission Category	Advertised for Spring 26-27	Broad Research Area	Remarks
TA	Yes	Research Areas mentioned in **L.4.**	-
PA	No	Not Applicable	-
FA	Yes	Research Areas mentioned in **L.4.**	-
IS	No	Not Applicable	-
EX	Yes	Research Areas mentioned in **L.4.**	-
CT	Yes	Research Areas mentioned in **L.4./2(B)**	-

For details **refer** Section B.

L.2.3. Interview

Only eligible applicants are permitted to participate in the selection process. The department will conduct two rounds of interviews for the shortlisted candidates. Both rounds will be conducted online. The first round of interviews (R1) is compulsory for all shortlisted candidates. After the first round of interviews, the department will call suitable candidate(s) for the second round of interviews (R2). The interview date and time for R1 and R2 will be emailed only to the shortlisted candidate(s). Applicants are advised to check the website regularly from time to time.

L.3. Syllabus for Interview

Candidates are expected to review the latest GATE Physics syllabus apart from general research aptitude questions.

L.4. Domains of Research

PhD positions in the following three research domains are available in the Department of Physics during the Spring 2026-27 semester. Applicants have to choose at least one of these topics and mention those in the application form with appropriate order of preference.

1. Experimental Condensed Matter Physics and Material Science

(A) Experimental Condensed Matter Physics (Dr. Santosh Kumar)
Application category to apply (refer Section B) – FA / TA / EX

Eligible social categories to apply
FA- Open to all social categories; TA- Open to OBC, ST, and SC.
Candidates should be motivated to work in the area of experimental Condensed Matter Physics. They should have some fundamental knowledge in the topics- Superconductivity and Magnetism.

Research Topics-

- **Superconductivity**- Vortex dynamics, vortex phase transitions and phase diagrams in a variety of superconductors.

- **Magnetism**- Magnetic anisotropy, torque magnetometry studies in some rare-earth transition metal based permanent magnets.
- **Single crystal growth**- Crystal growth of superconducting materials, rare-earth transition metal based permanent magnets and rare-earth free magnets.

The research work will involve advanced experimental studies of the magnetic, electrical, and thermodynamic properties of superconductors and permanent magnets using a Physical Property Measurement System (PPMS) at IIT Dharwad. The PPMS is a versatile, automated system used for investigating these properties over a wide range of temperatures and magnetic fields.

Collaboration- The research work will also be carried out in collaboration with University of Warwick UK, Indian Institute of Technology Bombay (IIT Bombay) and Tata Institute of Fundamental Research (TIFR), Mumbai, India. The experiments in IIT Dharwad will be performed in high quality single crystals grown at University of Warwick, UK. In addition, the students will be given the opportunity to visit and perform experiments at IIT Bombay and TIFR Mumbai.

(B) Experimental Material Science (Dr. Dhriti Sundar Ghosh)

Application category to apply (refer Section B) – FA, TA

FA- Open to all social categories; TA- Open to ST/SC only

We invite motivated candidates to join our research group and contribute to the development of next-generation optoelectronic devices such as perovskite solar cells, photodetectors, LEDs, etc. Our focus is on creating high-efficiency, lightweight, and stable devices through advanced thin-film deposition, interface engineering, and scalable device integration.

Research Focus

- Thin-Film Fundamentals & Processing
    - Fabrication, characterization, and tailoring for optoelectronic devices
- Perovskite Optoelectronics
    - Modeling, materials, devices, testing, and integration

Candidate Role

You will work on:

- Fabrication, characterization, and optimization of thin-film and perovskite device architectures
- Techniques including physical vapor deposition (PVD), spin-coating, and advanced spectroscopy tools
- Collaborative projects with national and international partners in photovoltaics and materials science
- Device modeling and simulation (programming knowledge is an advantage)

What We Offer

- Access to state-of-the-art thin-film, spectroscopy, and photovoltaic characterization facilities
- Opportunities to publish in high-impact journals and present at international conferences
- A collaborative environment within a rapidly growing photovoltaics research group
- For details visit: **www.thinpvlab.in**

2. Photonics and Quantum Information

(A) Photonics (Dr. Kavita Devi)

Application category to apply (refer Section B) – FA / TA / EX

FA- Open to all social categories; **TA/EX-** Open to OBC, SC, and ST only
Candidate should be highly motivated to work in the field of Laser Technology and Nonlinear Optics with the focus on experimental design/development/characterization of nonlinear frequency conversion/optical parametric oscillators/spontaneous parametric down conversion/optical devices/optical materials, together with theoretical modelling and simulations. Experimental working experience together with computer simulation skills, programming language (MATLAB, Python, C++, CUDA), graphing and analysis (Origin) abilities are desirable.
The Ph.D. position in the area of Photonics will provide the candidate with:

- The opportunity to work in the state-of-the-art Photonics Laboratory equipped with sophisticated equipment and facilities
- Publications in international journals and participation in international conferences.
- Collaboration with research groups across the globe.
- Industrial exposure.
- For details visit the NLOP Group webpage.

Topic - Optical devices based on nonlinear frequency conversion across unprecedented wavelengths in all time scales; Optical Frequency Combs; Dynamics of optical parametric oscillators; Photo-physical response, and investigation of various optical nonlinear materials.

(B) Quantum Information Theory (Dr. R. Prabhu)

Application category to apply (refer Section B) – TA / FA

Eligible social categories to apply: FA - Open to all social categories; TA - Open to OBC, ST, and SC only
Candidates applying for this research area should be highly motivated to conduct theoretical research on topics at the intersection of quantum mechanics, quantum optics, many-body physics, and relativity. There are also opportunities to explore cutting-edge Quantum Technologies, including Quantum Computation, Quantum Communication, Quantum Metrology, and Quantum Artificial Intelligence. A solid understanding of the foundations of quantum mechanics, quantum field theory, and quantum information science is essential. Additionally, candidates should be proficient in executing advanced computational programs using any of the standard programming languages. Knowledge of the key principles in these fields, coupled with strong analytical and programming skills, will be highly advantageous.
Topic - Quantum information theory; its interface with quantum optics, relativity, and many-body physics; Quantum Communication; Quantum Computation, Quantum Metrology, Quantum Artificial Intelligence.

3. Atomic, Molecular and Optical Physics, Remote Sensing

Application category to apply (refer Section B) – FA / TA / EX

Eligible social categories to apply – GEN, OBC-NCL, SC, and ST under TA/EX are considered; For FA category vacancies are open to all social categories.

(A) Atomic, Molecular and Optical Physics (Dr. Koushik Saha)

The candidate should be highly motivated to pursue research in the area of Experimental Atomic and Molecular Physics. The research work is related to studies of molecular dynamics in ultrafast timescales. The work to be carried out is experimental in nature. A good background in Atomic, Molecular and Optical Physics and a strong aptitude for experimental work is necessary. Experience in working with lasers, high vacuum generation, time of flight mass spectroscopy etc. together with computer simulation skills, MATLAB programming are desirable. The Ph.D. position provides the candidate with unique prospect to develop and work with state of the art experimental facilities, performing experiments using femtosecond lasers and advanced setups such as RIMS, COLTRIMS and VMI as well as theoretical modeling and simulations. The candidate will also have the opportunity for collaborative research work with labs of global repute.

Topic - Experimental Atomic and Molecular Physics

- Molecular dissociation dynamics due to dissociative ionization brought about the electron beams and lasers.
- Suppression or enhancement of ionization/fragmentation of molecules due to different excitation regimes by various ionizing agents.
- Strong field ionization and dissociation of molecules by intense femtosecond laser pulses.

Appendix A: Sponsorship Certificate for Ph.D. External Registration (EX)

(To be typed on letterhead of the Sponsoring Organization)

i. Name of the applicant:
ii. Name of the sponsoring organization:
iii. Address:
iv. Present Designation of the Applicant:
v. Present status of the applicant: (Permanent/Semi-permanent/Temporary)
vi. Division where research work is proposed to be done:
vii. Name of supervisor from the sponsoring organization:
viii. (Biodata of the supervisor to be enclosed giving details of designation, qualification, research experience, etc.)
ix. Details of facilities relevant to the research problem will be made available to the candidate by the organization.

Statement of proposed Co-supervisor (external)

If Shri / Kum. / Smt._____
is registered for the doctorate degree, I,_____,
agree to act as his/ her research Co-supervisor along with the research Supervisor from IIT Dharwad.

Date: Signature of proposed Co-supervisor (external)

Statement of sponsoring authority

If Shri. /Kum. / Smt. _____

is admitted to the Ph.D. program, we shall allow him/ her to undergo the program of studies at IIT Dharwad.

Further, we shall fully relieve him/her from normal duties to complete the course work requirement (and qualifier examination, if applicable) at IIT Dharwad.

During the period of Doctoral program, the candidate will be permitted to carry out his / Her research work at our laboratories / organization and will be given the required facilities.

We also give our consent to Shri. /Kum. / Smt./Dr._____
of our organization to be the Co-supervisor (external) of the Ph.D. thesis, along with a faculty member of IIT Dharwad as the Supervisor.

Date: Signature and Seal of the Sponsoring Authority

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